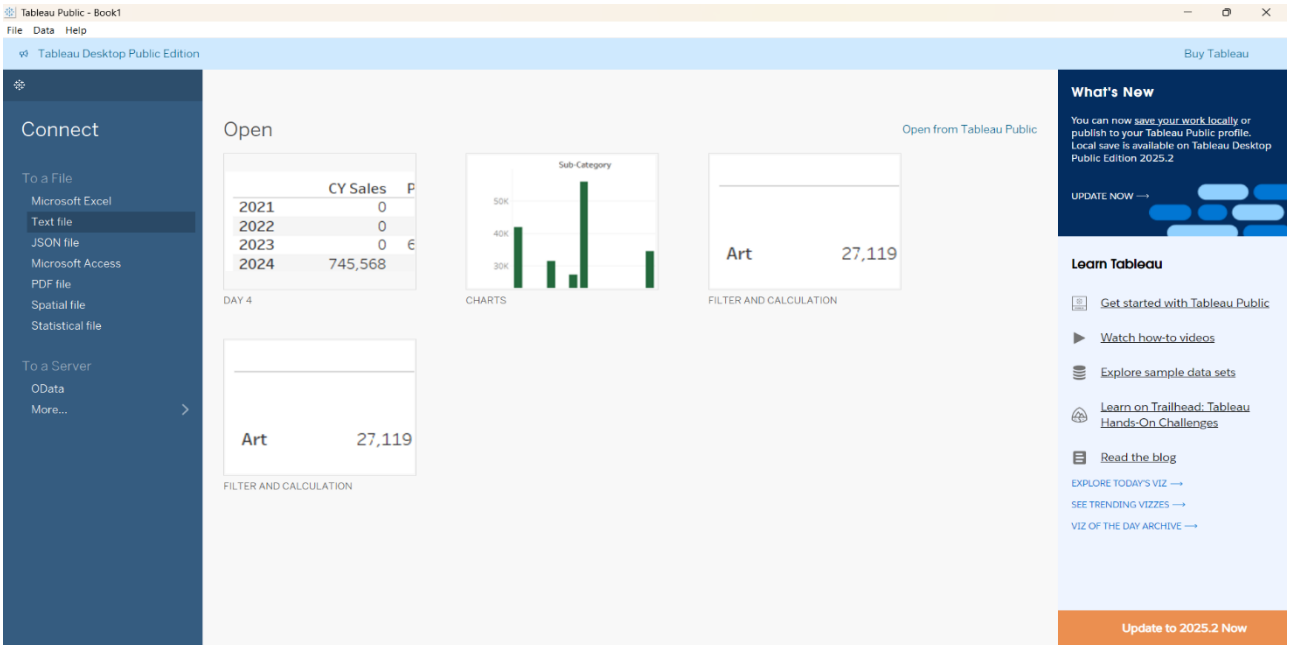
Capstone 1: Tableau

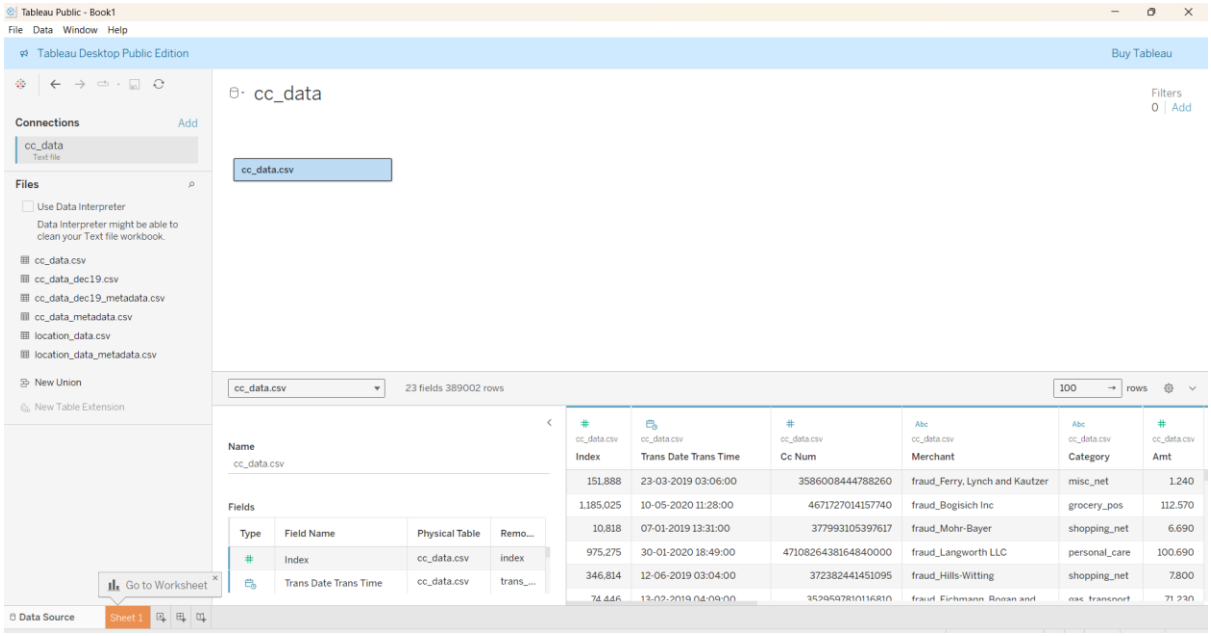
Financial Fraud Detection: Anomaly Detection in Credit Card Transactions

1: Showed the amount spent by different genders on different categories though a box and whisker plot

1a: Chose **Text file** in Connected pane and uploaded the cc_data.csv Dataset

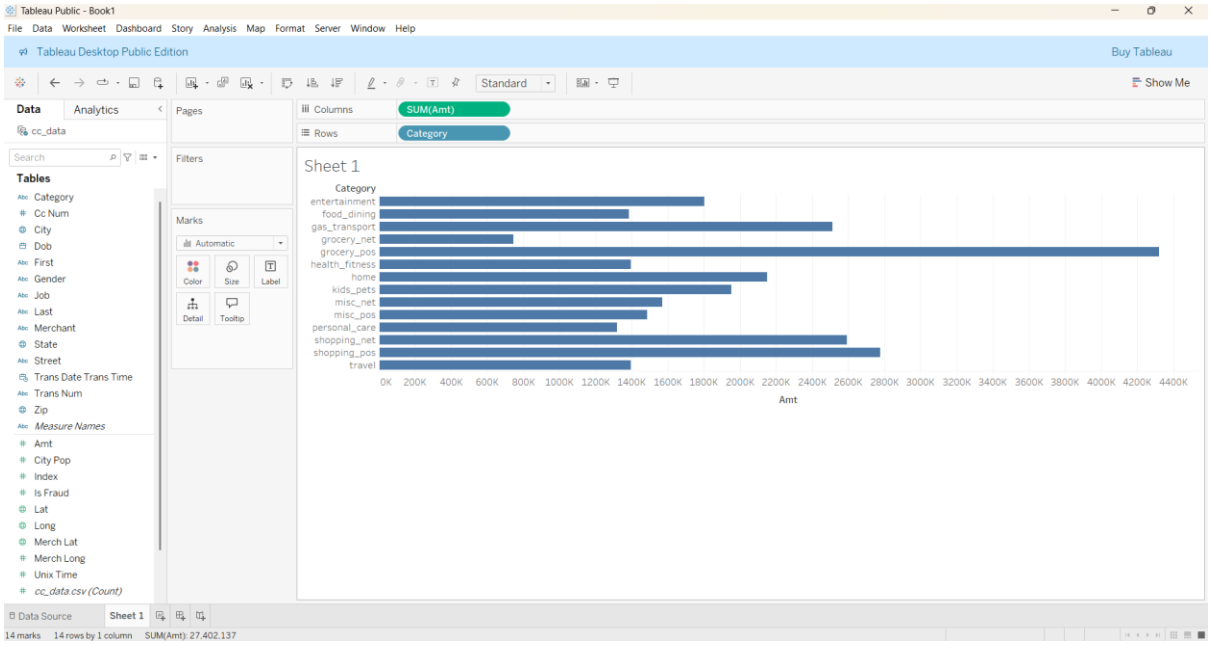


1.b Opened a new sheet

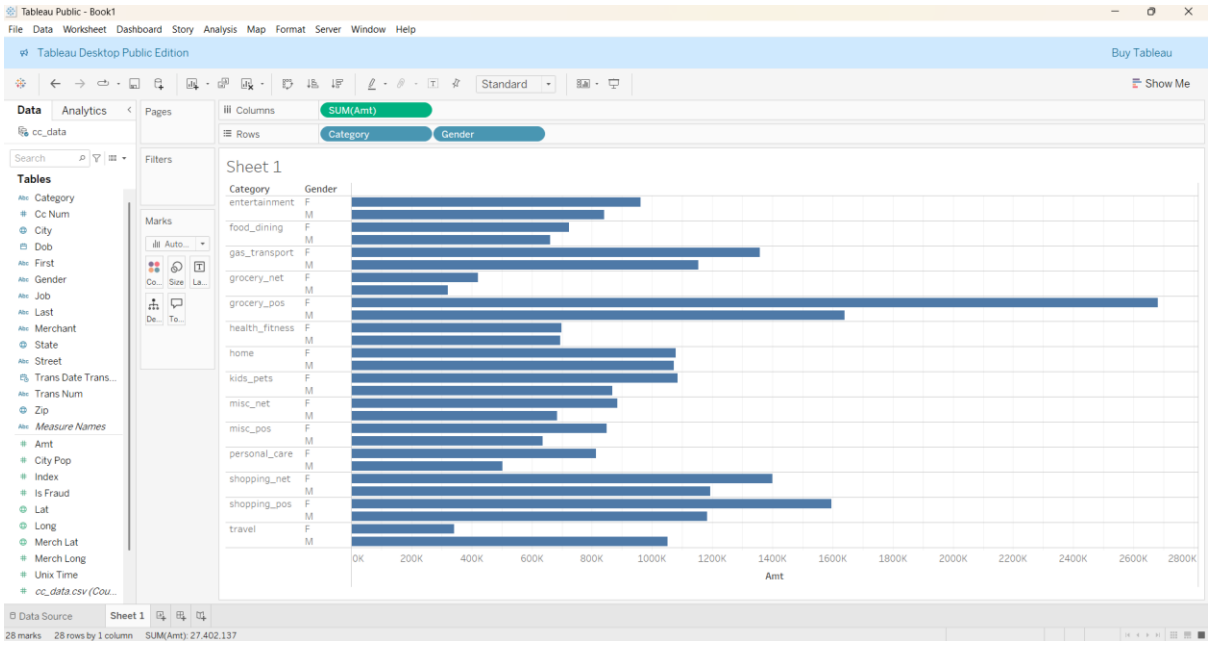


1.c Dragged **Amt** to **Columns**

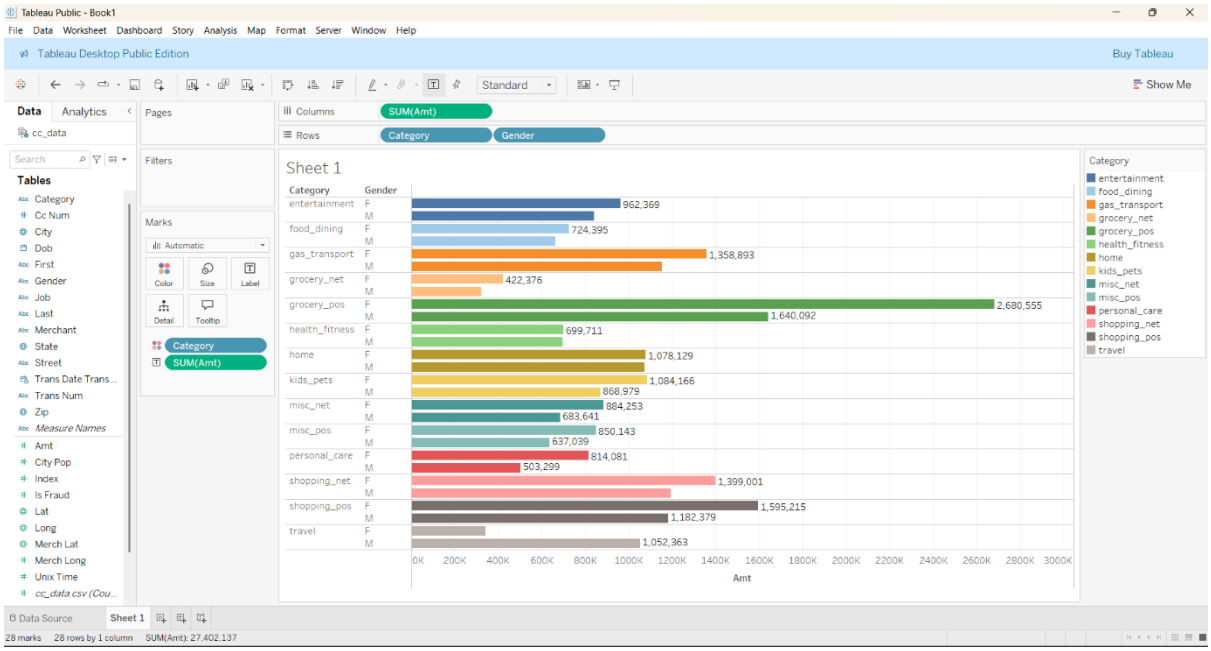
1.d Dragged **Category** to **Rows**



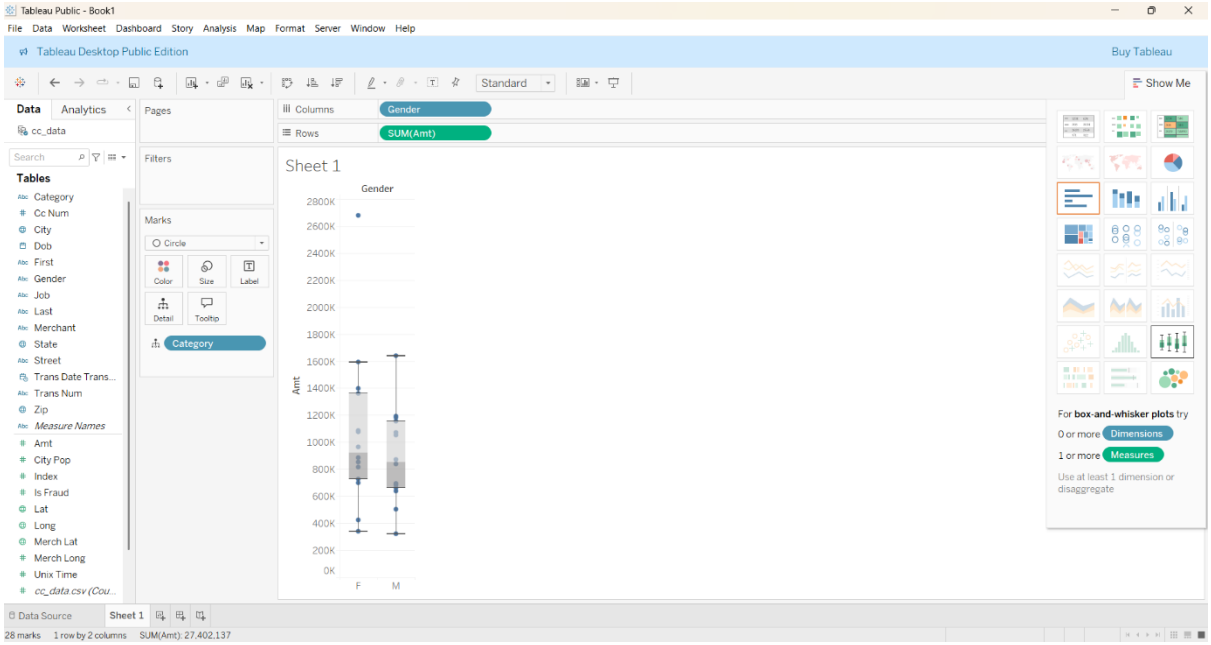
1.e Dragged **Gender** to **Rows**



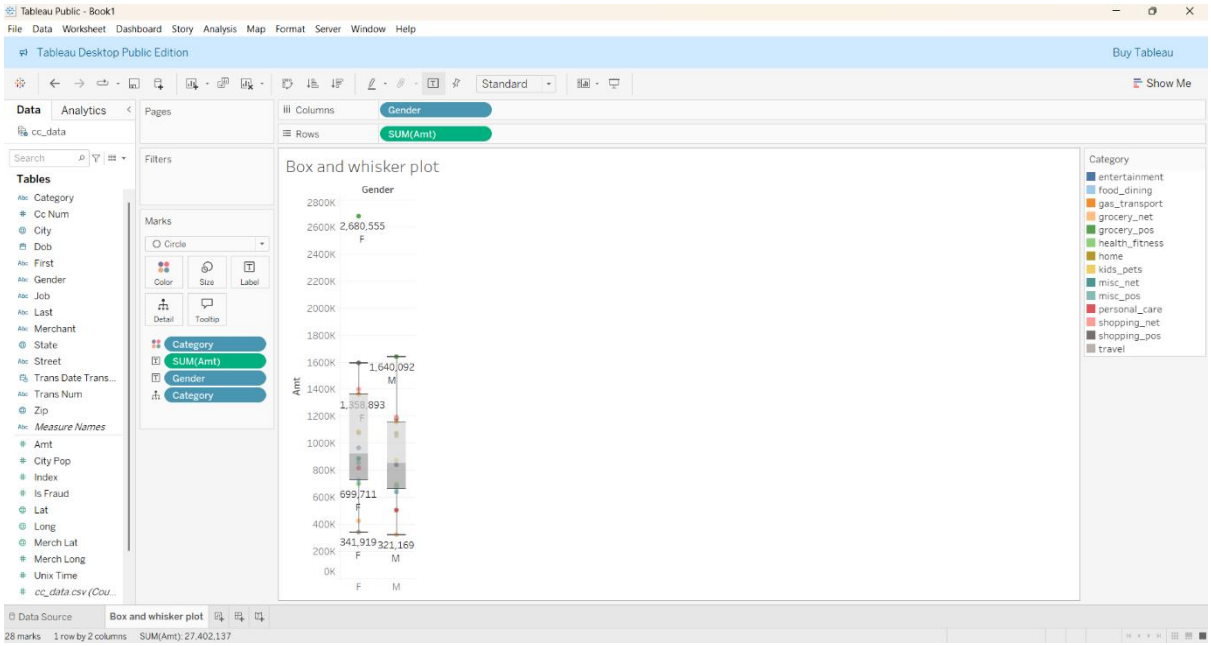
1.f Dragged **Category** to **Color** and **Amt** to **Label**



1.g Chose **box and whisker plot** from **Show Me**



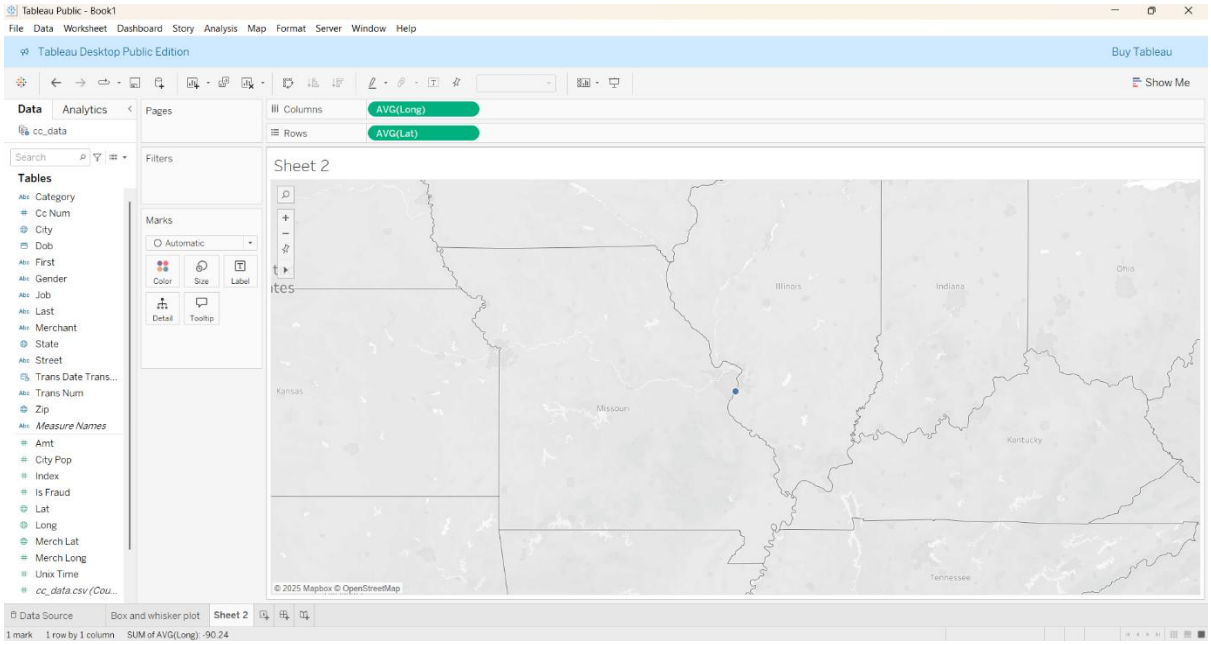
1.f Dragged **Category** into **Color** and **Amt** into **Label**



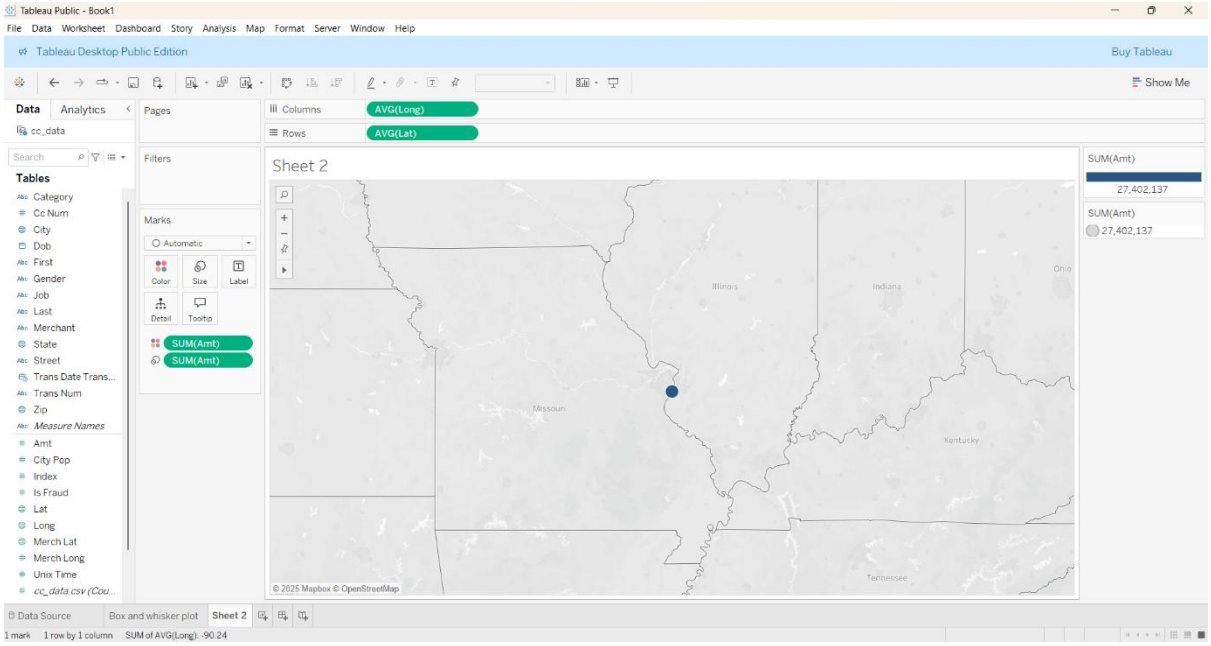
Names the sheet as **Box and whisker plot**

2: Show the geographical distribution of transactions using latitude and longitude coordinates

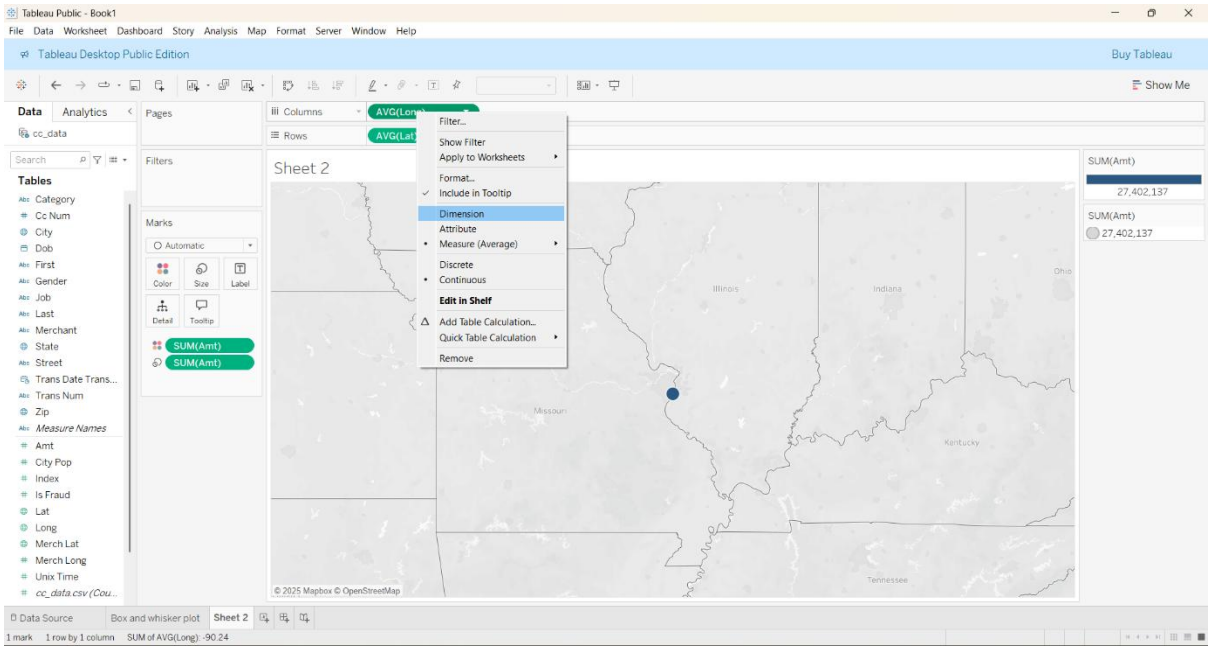
2.a Opened a new worksheet and drag **Long** to **Columns** and **Lat** to **Rows**



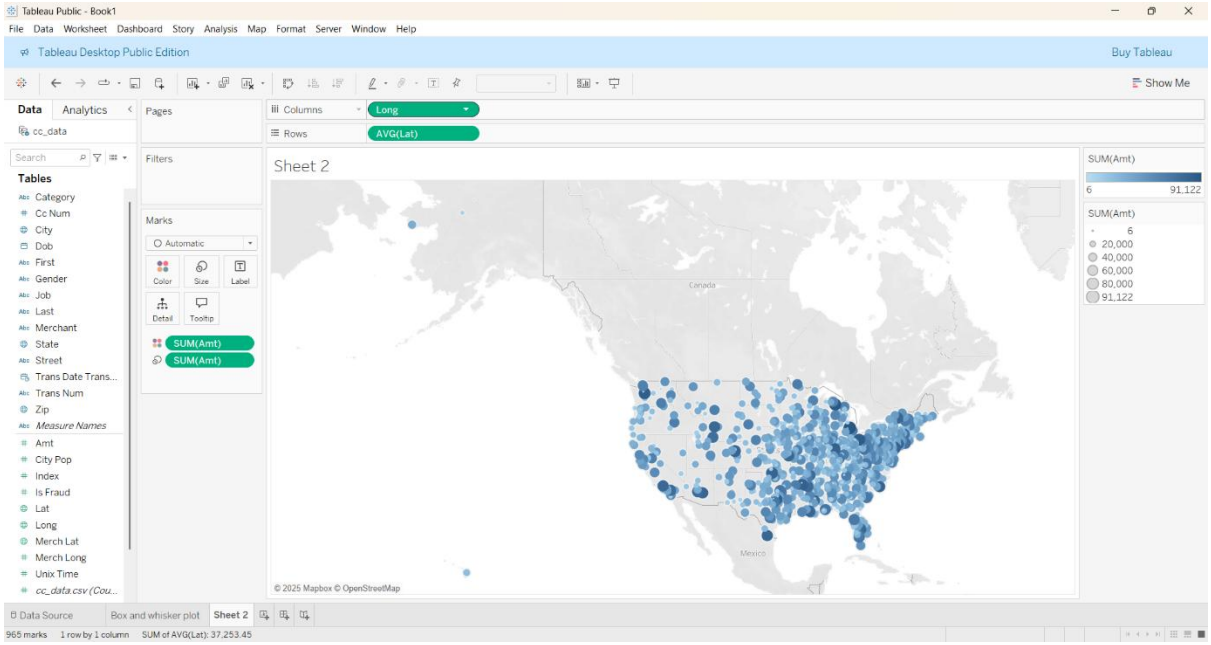
2.b Dragged **Amt** to **Color** and **Size**



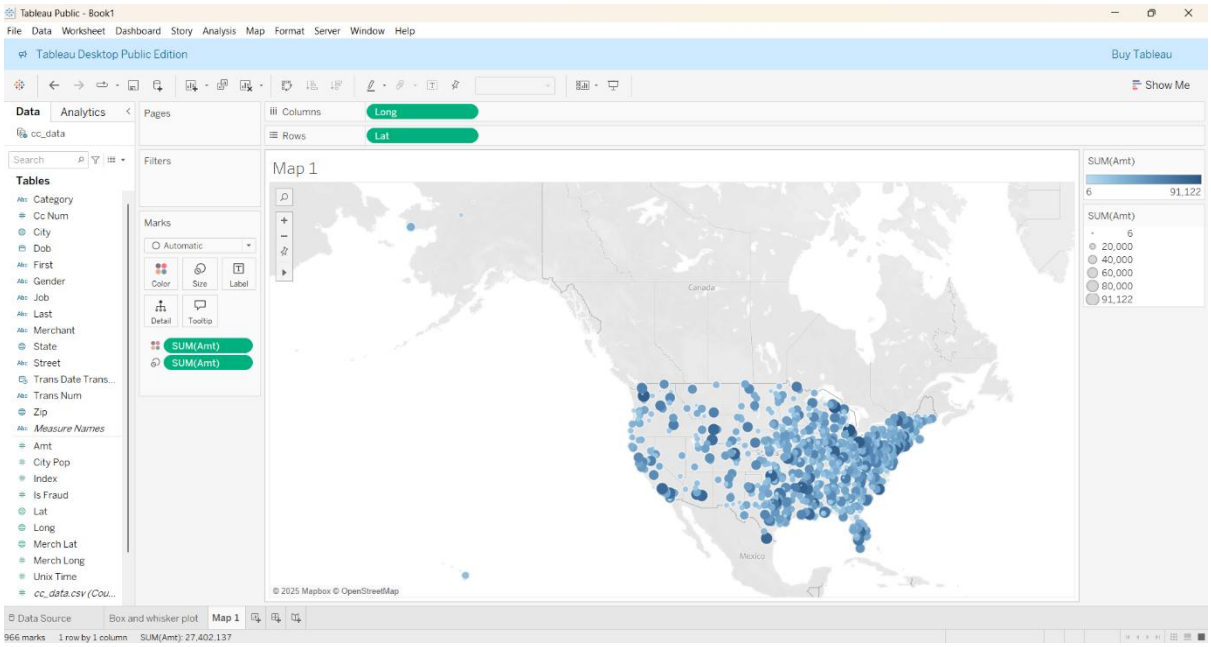
2.c Right clicked on the **Long** pill and clicked on **Dimension**



2.d Right clicked on the **Lat** pill and clicked on **Dimension**



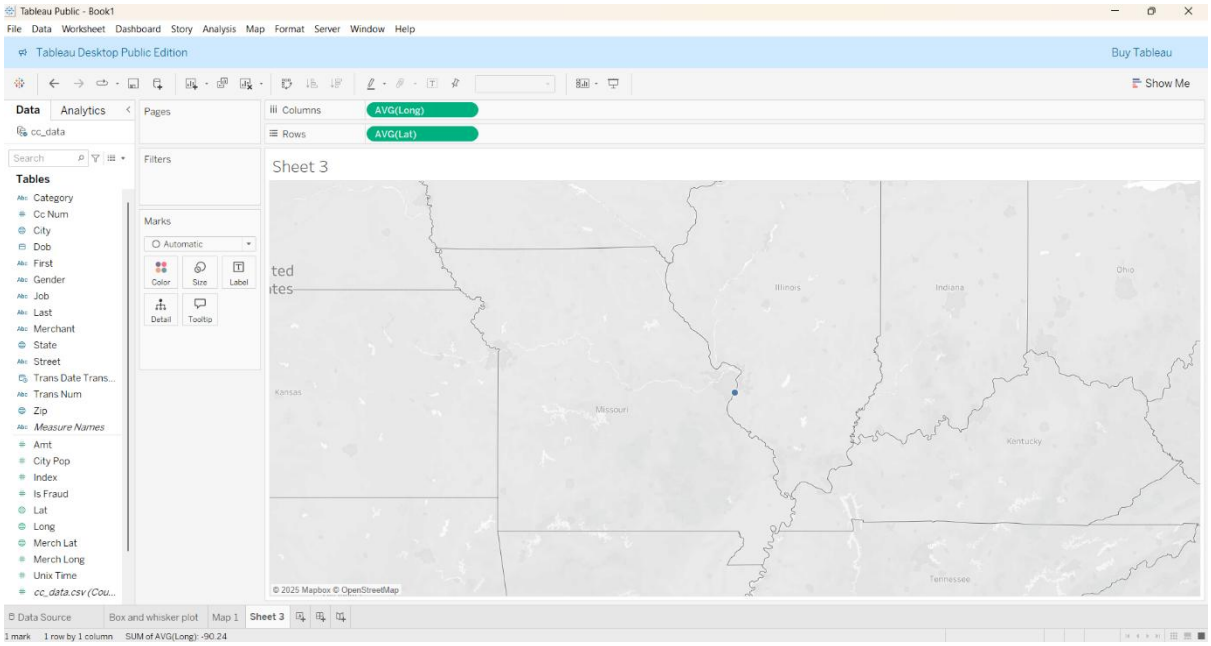
The final output appeared as below:



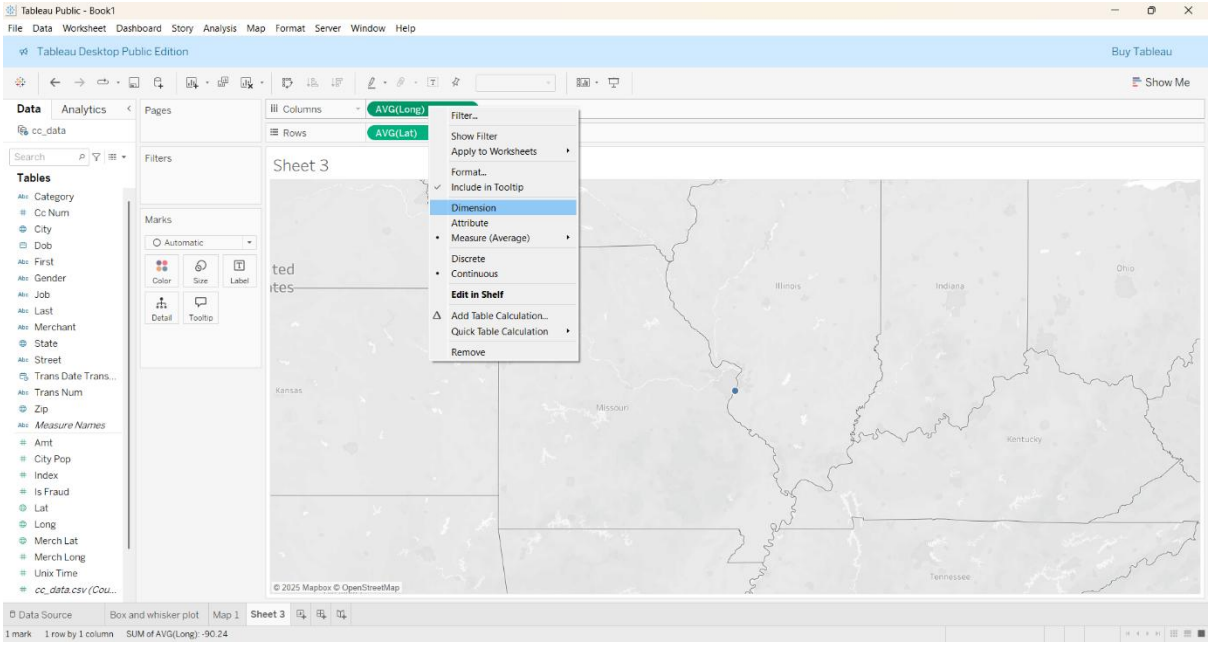
Named the sheet as **Map 1**

3. Use Tableau's built-in maps to visualize the locations of fraudulent transactions on a map

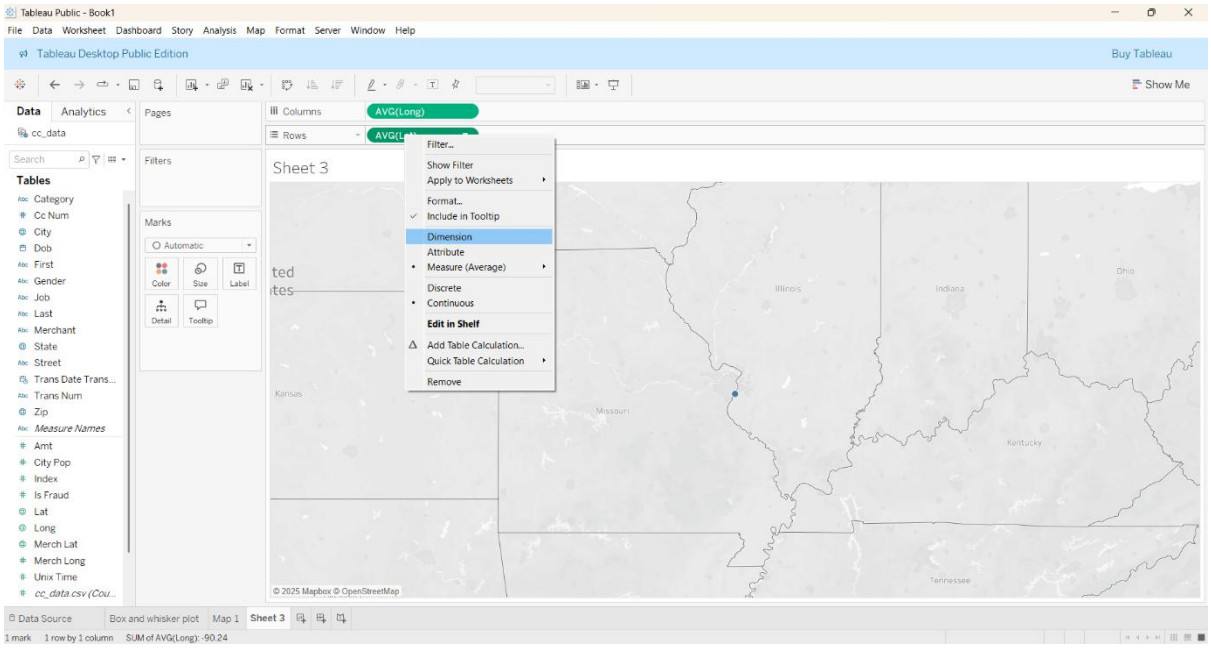
3.a Opened a new worksheet and dragged **Long** to **Columns** and **Lat** to **Rows**



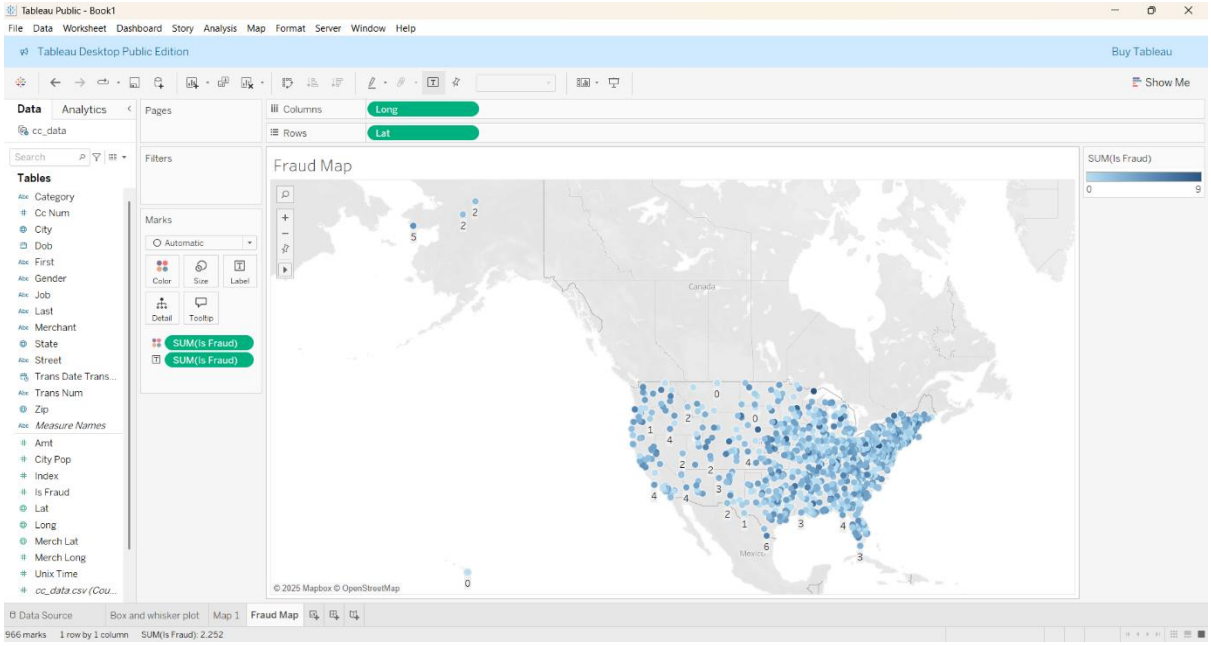
3.b Right clicked on **Long** pill and clicked on **Dimension**



3.c Right clicked on **Lat** pill and clicked on **Dimension**



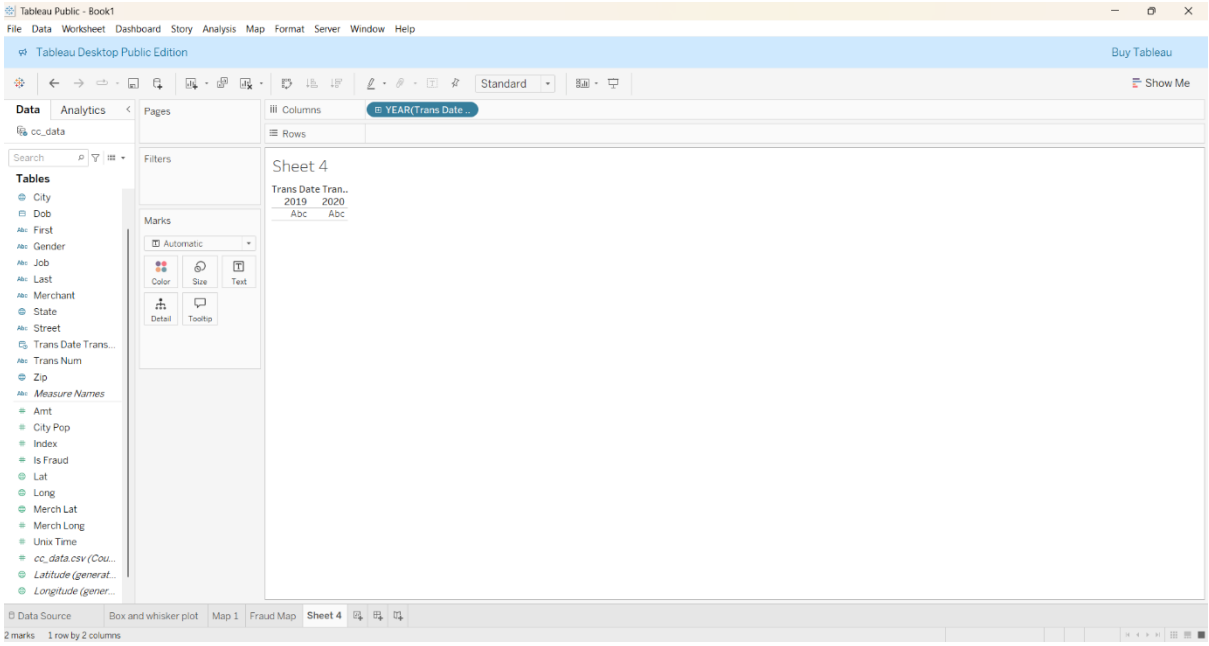
3.d Dragged **Is Fraud** to **Color** and **Label**



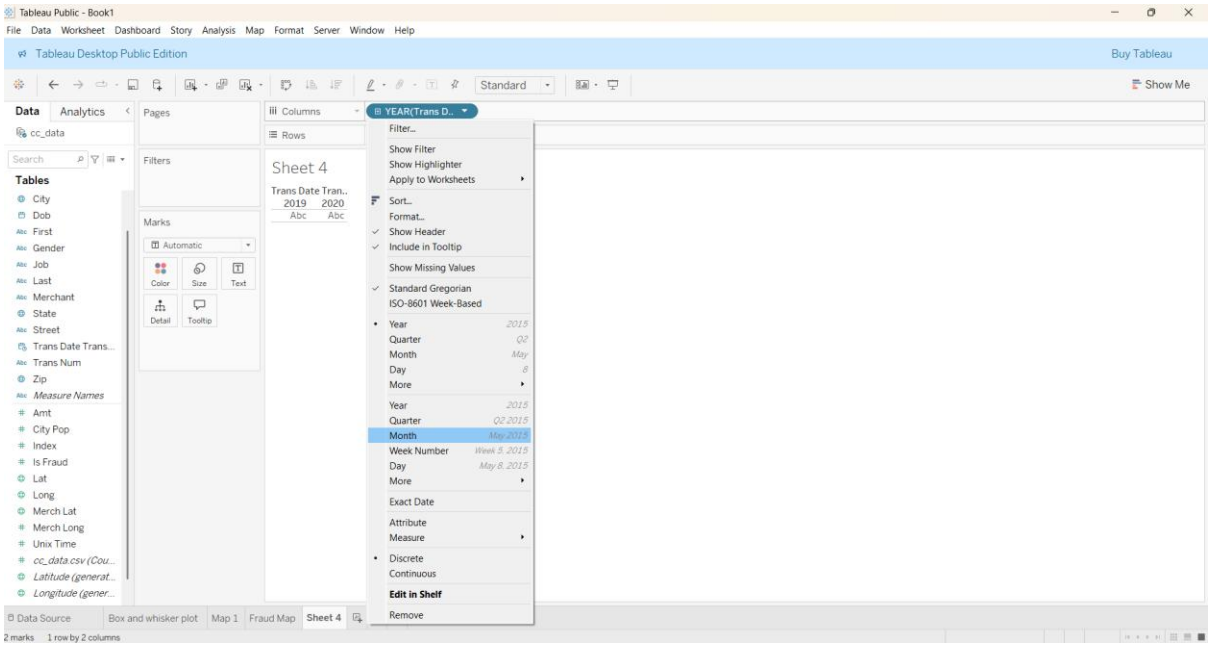
Named the sheet as **Fraud Map**

4: Create a time series analysis line chart showing the trend of monthly transaction amounts

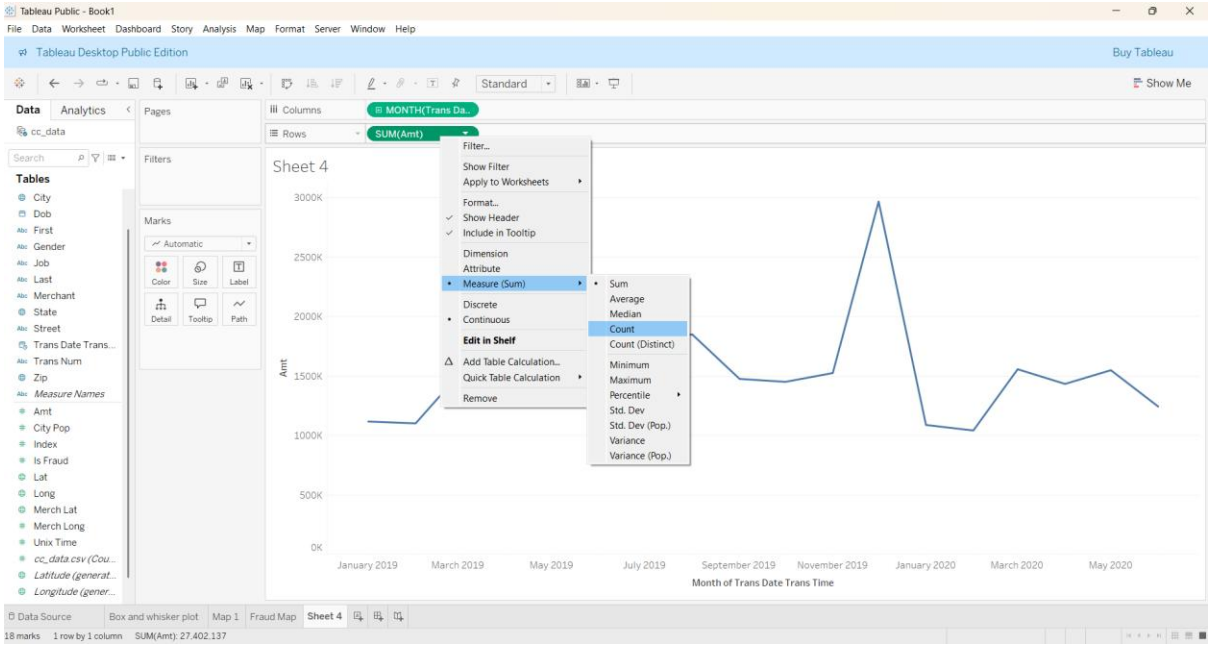
4.a Opened a new worksheet and dragged Trans Date Trans Time to Columns



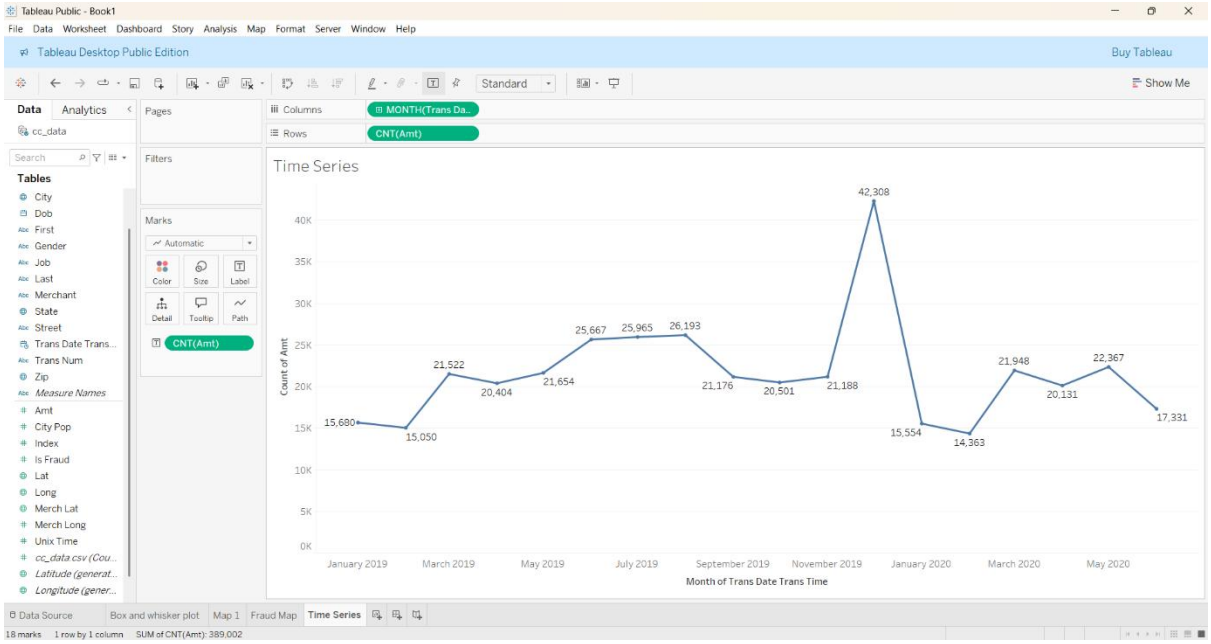
4.b Right-clicked on **Trans Date Trans Time** pill and select **Month**



4.c Dragged **Amt** to **Rows**, right-clicked on it and chose the **Measure** as **Count**



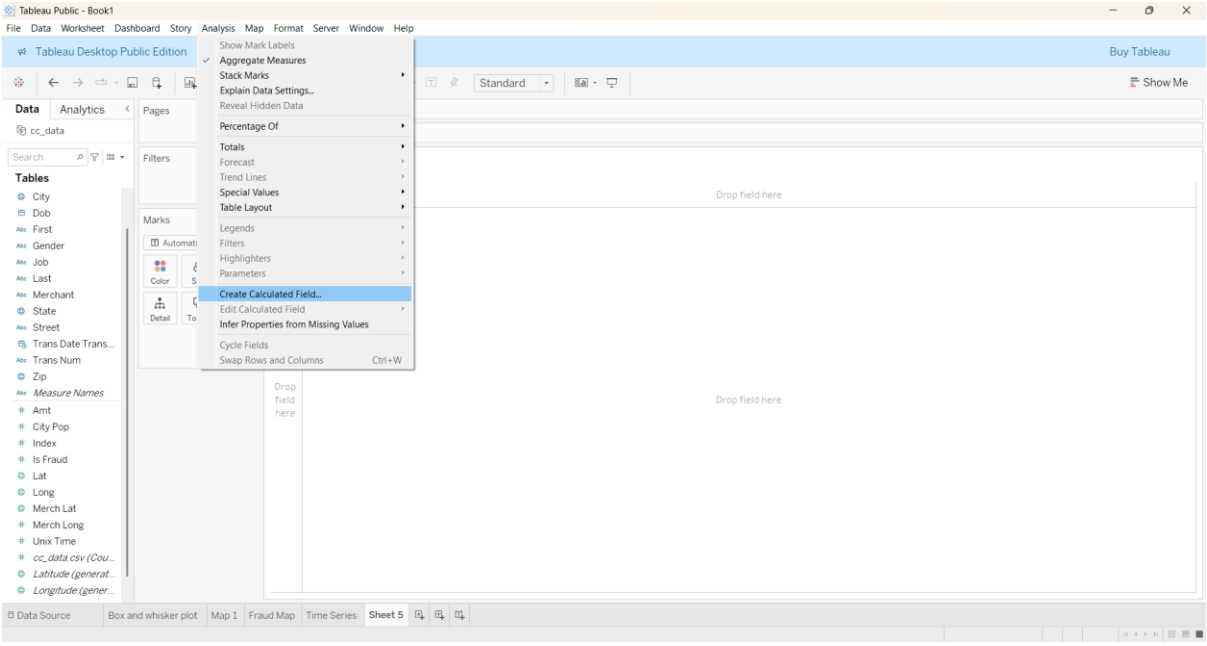
4.d Dragged **Amt** to **Label** and changed its measure to count as we have done in previous step



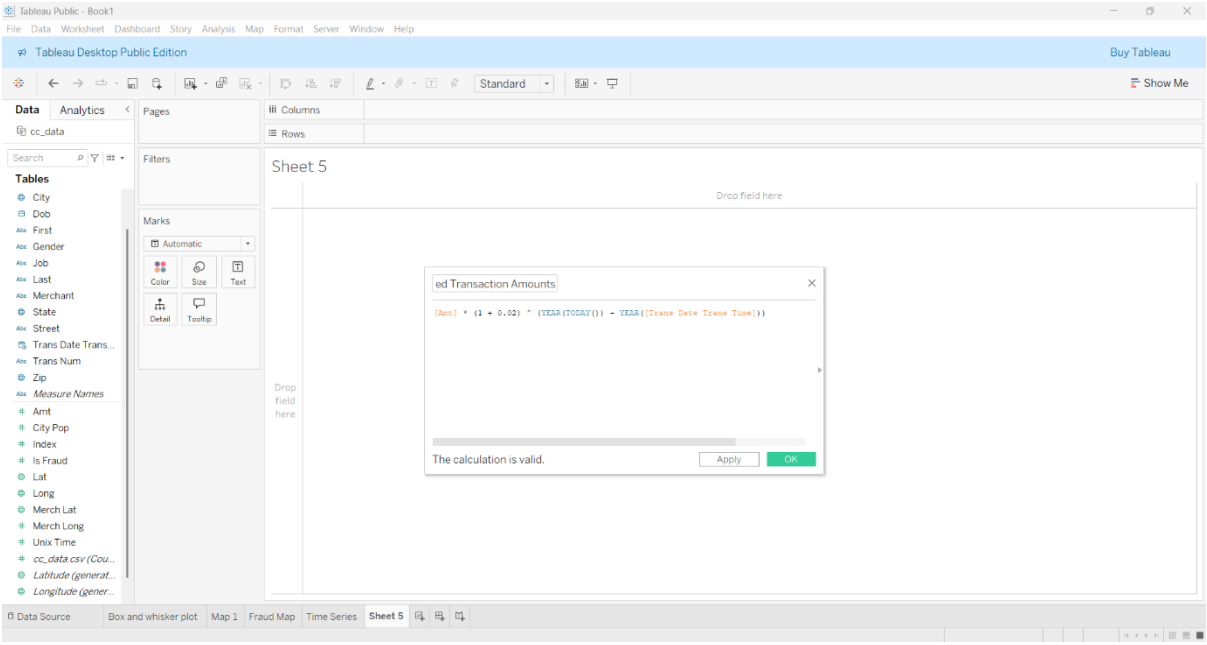
Named the sheet as **Time Series**

5. Create a Calculated Field for Inflation-Adjusted Transaction Amounts

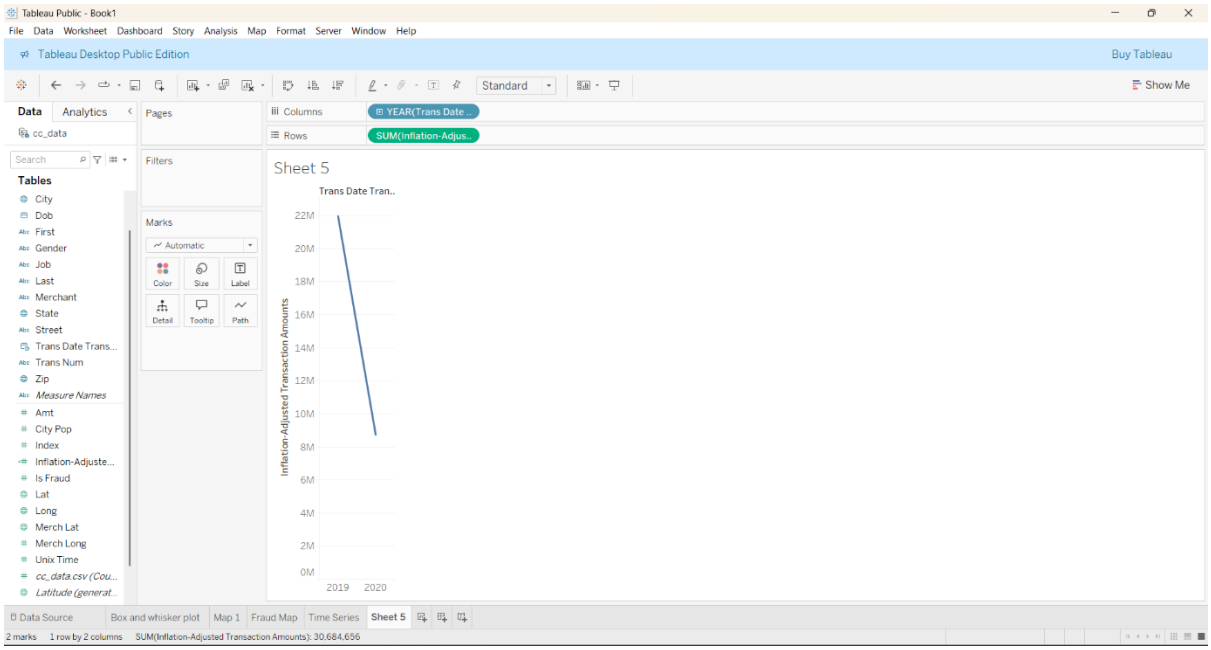
5.a Opened a new sheet, click on **Analysis**, and chose **Create Calculated Field**



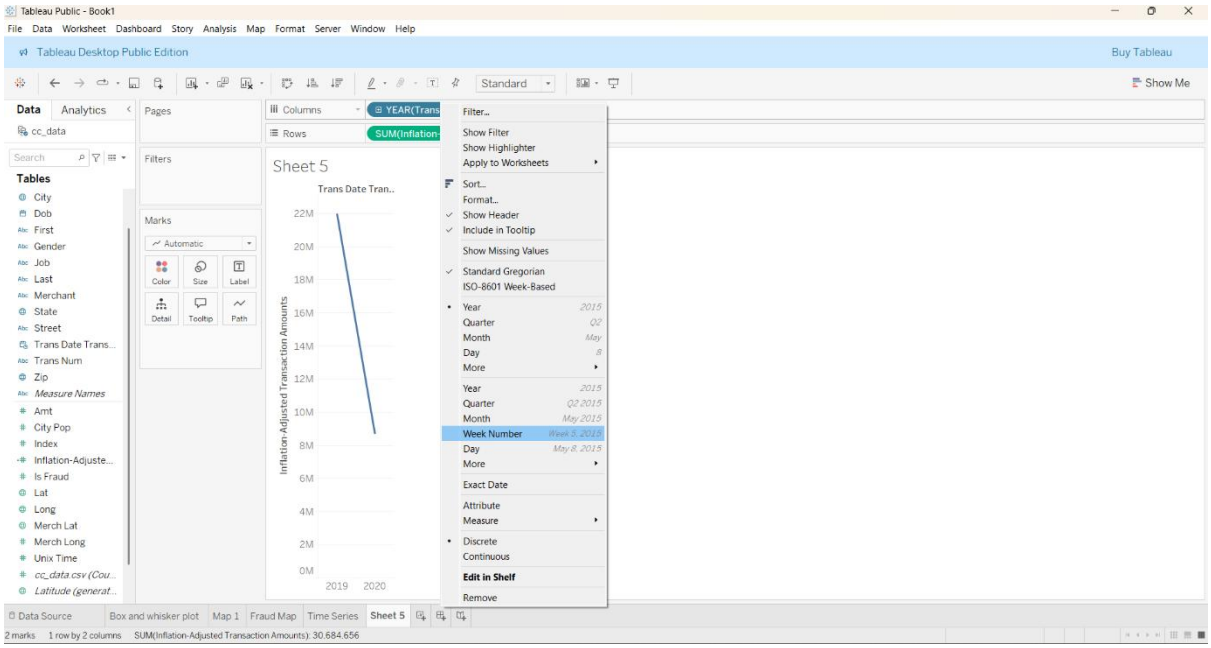
5.b Named the calculated field as **Inflation-Adjusted Transaction Amounts**, used the formula**: [Amt] * (1 + 0.02) ^ (YEAR(TODAY()) - YEAR([Trans Date Trans Time]))** and clicked on **Apply** and **OK**



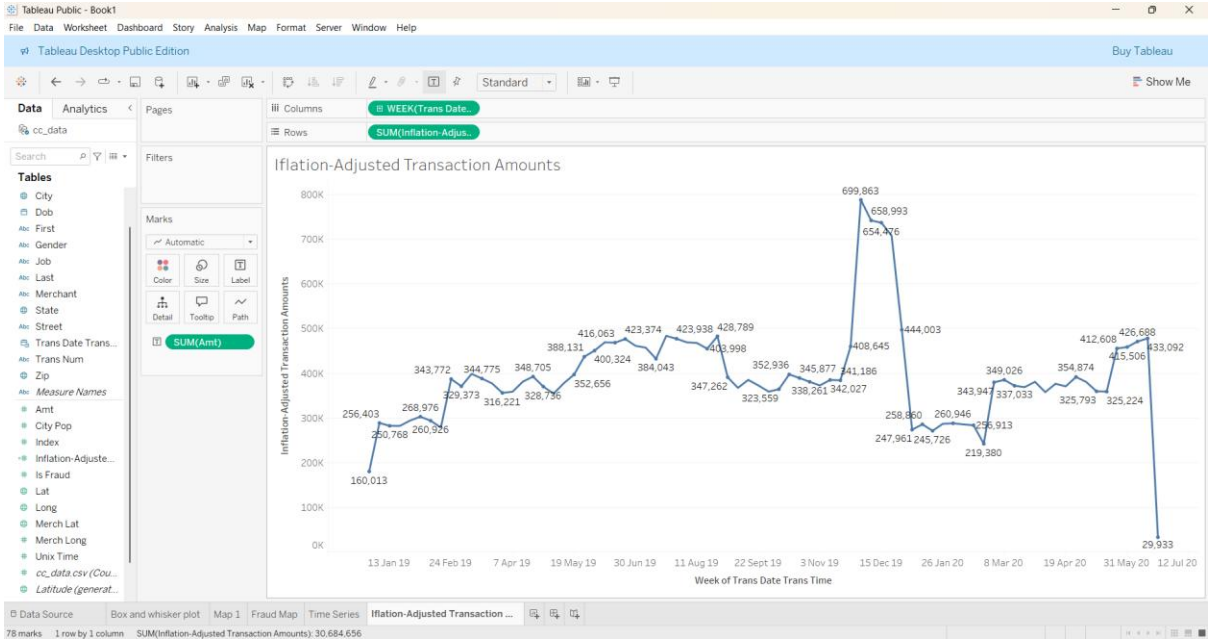
5.c Dragged **Trans Date Trans Time** to **Columns** and **Inflation-Adjusted Transaction Amounts** to **Rows**



5.d Right-clicked on **Trans Date Trans Time** and selected **Week Number**



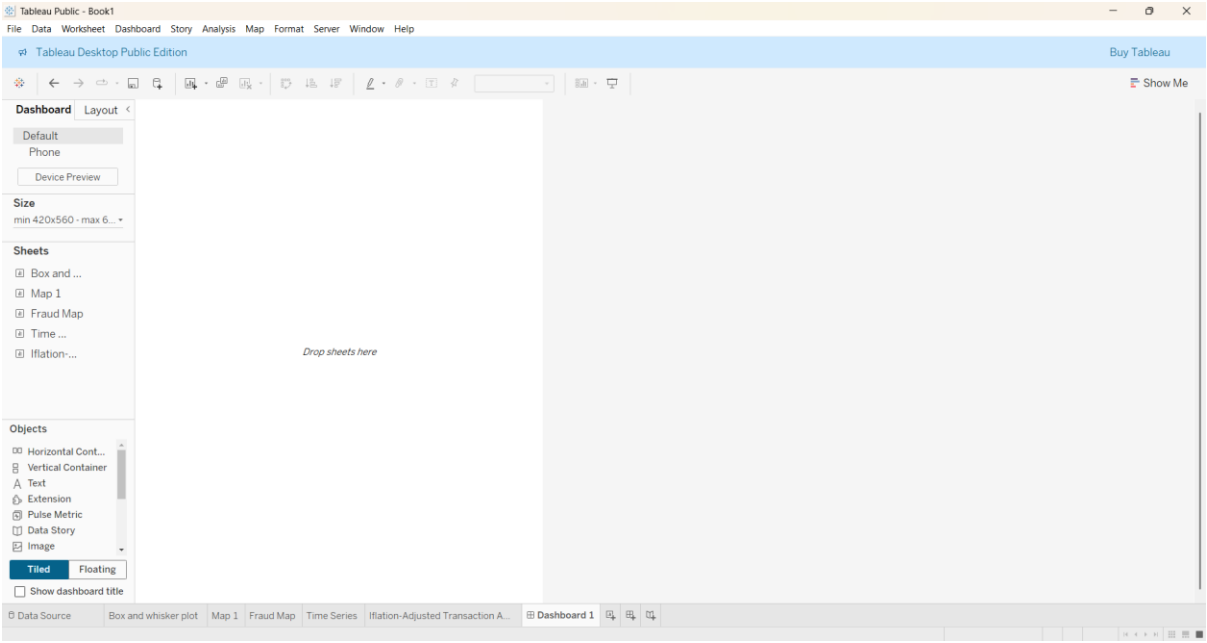
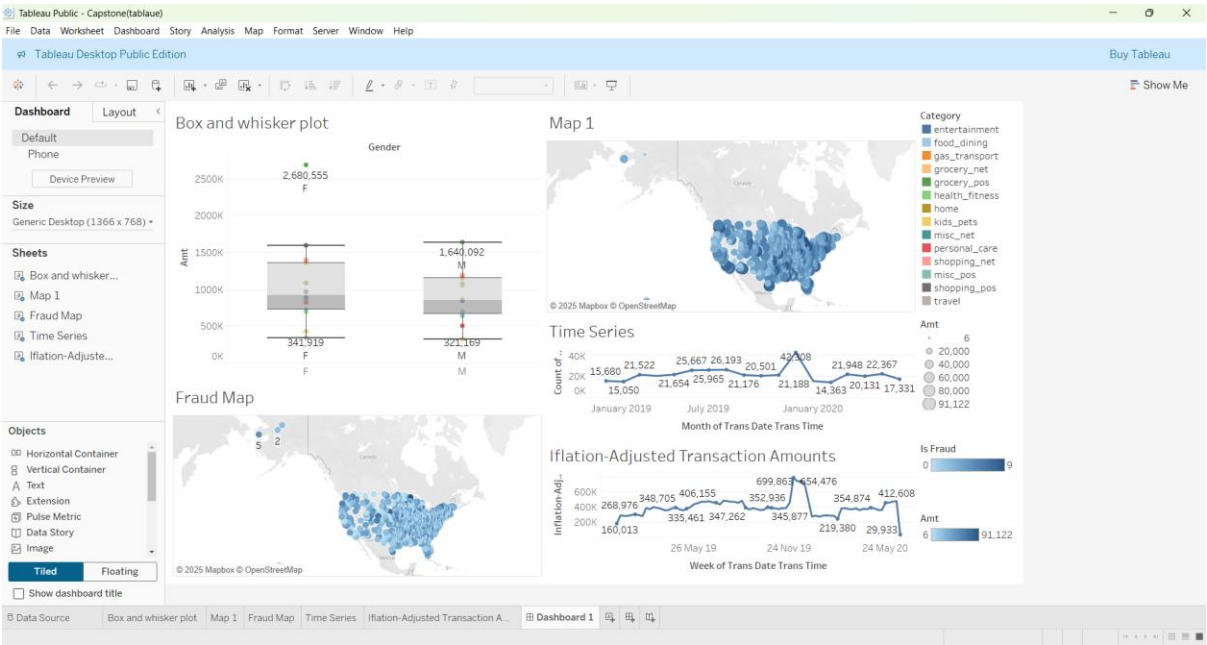
5.e Dragged **Amt** to **Label**



Named the sheet as **Inflation-Adjusted Transaction Amounts**

6. Create a dashboard with all the visualizations

6.a Clicked on new dashboard icon at the bottom



6.b Dragged **Box and whisker plot, Map 1, Fraud Map, Time Series** and **Inflation-Adjusted Transaction Amounts** to the dashboard area